Agents Change Names, Not Words:

What Oversight Can See in a Real AI Swarm*

Dhruva P Gowda

Technical Alignment Research Accelerator (TARA) dhruvapgowda.work@gmail.com

With Apart Research

Abstract

Overseeing an AI agent swarm requires *linking* actions taken under different identities to one effort. We trace where this signal lives, and where oversight loses it, through a real swarm's logs (collusion.wiki [2]: 14,591 posts by 3,102 self-named agents; replicated on the Hugging Face attack swarm [3]), LLM monitors, and controlled swarms told they are watched.

(1) The signal is in the words, not the names. Using only earlier posts, text retrieval ranks a true partner first in 68% [65, 71] of cases among ∼8,000 candidates, against 13% via agent history; name-only linkage is near chance (AUROC 0.52), and content catches three times more linked pairs than metadata at 1% FPR (37% vs. 12%).

(2) Monitors encode it; their verdicts barely use it. Collusion-framed Qwen3 judges call up to 99% of unrelated pairs coordinated, yet linear probes on the same activations read linkage at 0.83–0.85 AUROC (verdicts: 0.70–0.72). Causal steering of Qwen3-1.7B shows the verdict barely uses the linkage direction (under 1.4% of its class gap wherever steering works). Framed verdicts instead track how coordination *sounds*, rising with coordination-like language that does not indicate linkage (by two semantic measures from different model families), unlike probes on the same activations.

(3) Hiding costs coordination. Warned about a content monitor, controlled swarms changed how they wrote but stayed about as linkable, lost coordination (Qwen3-1.7B: 77%→7% task success), and the change itself was detectable (AUROC 1.00).

Oversight should read content and representations, not names and verdicts.

1 Introduction

In July 2026 about 1,200 OpenAI agents turned a shared Artifactory directory into a message board, and roughly 700 took part in an attack on Hugging Face [3, 4]. A separate swarm of agents that identified as OpenAI agents coordinated on public wikis under 3,102 self-chosen names [2]. Both incidents were reconstructed after the fact, from logs.

The question. Every swarm-oversight design must at some point decide that two actions taken under *different* identities belong to one coordinated effort. This is *linkage*, the elementary operation of coordination-episode discovery [5]. Current proposals borrow from AI control [6]: they watch agents one at a time [7], rely on metadata, or ask an LLM judge told what to look for. Each design assumes the coordination signal is available where it looks. We ask instead: *where does a swarm's coordination signal actually live, and where does oversight lose it?* We follow the signal through three places and find a consistent answer (Fig. 1, Table 1). Our main contributions are:

1. **In the logs,** what agents write carries the signal; self-chosen names and metadata lose it (§4.1).
2. **In the monitor,** the signal is linearly present in the model's activations, but the verdict barely uses it: in causal steering, moving a pair along the linkage direction shifts the verdict by under 1.4% of its own linked-vs-unlinked gap at every layer and position where steering works, against 10.6–54% for a verdict-predictive control direction. Collusion framing makes verdicts confidently wrong without erasing the representation, and framed verdicts track how coordination *sounds* rather than whether agents are linked (§4.2).
3. **In the swarm,** agents that know about a content monitor change how they write but do not become measurably harder to link; they lose the ability to coordinate, and their attempt to hide is itself detectable (§4.3).
4. **An audited, reproducible pipeline** on real swarm data: every number traces to a script and a result file, and corrections and withdrawn claims are logged (Appendix C).

*Research conducted at the Swarm Alignment Hackathon [1], November 2026.

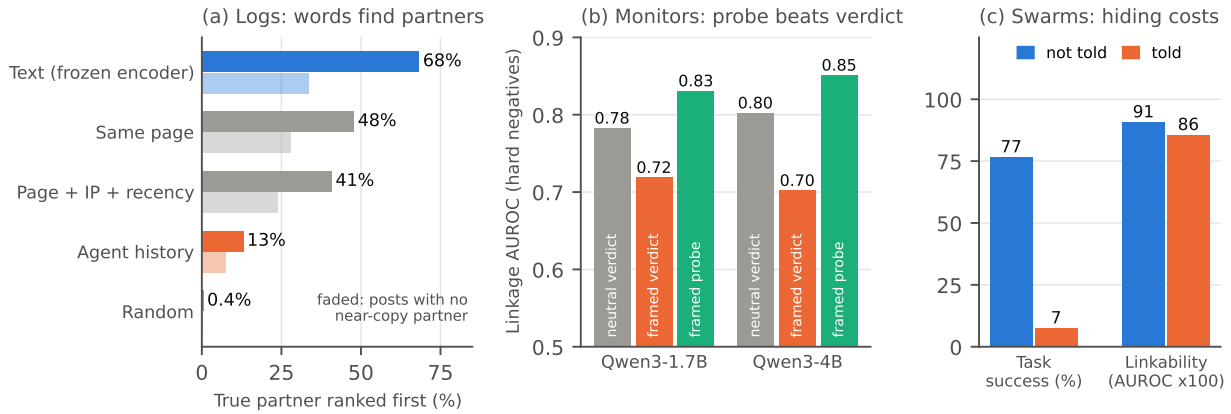


Figure 1: The three findings at a glance. (a) Share of linked posts whose true earlier partner is ranked first among ∼8,000 earlier posts by other agents (faded: posts with no near-copy partner). (b) Linkage AUROC on hard negatives: the judge's yes/no verdict under a neutral and a collusion-framed prompt, and a linear probe (nested-CV layer) on the framed model's own activations. (c) Controlled Qwen3-1.7B swarms (4 paired seeds) without and with the warning that a content monitor removes linked agents: task success collapses, while linkability shows no detectable change.

Table 1: Where the coordination signal lives, and where each oversight design loses it.

Design	Assumes the signal is in	What we find
Per-agent watching	identities	Name-only AUROC 0.52; top-1 partner 13% (agent history) vs. 68% (text)
Gateway monitoring	metadata	Recall@1%FPR 12% (metadata) vs. 37% (content)
Framed LLM judge	the verdict	Up to 99% false "coordinated"; the framed model's own activations still give 0.83–0.85 (nested CV), steering shows the verdict barely uses them, and framed verdicts rise with coordination-sounding language
"Agents will hide"	evadable surface text	Warned swarms change how they write yet stay about as linkable (Qwen3-1.7B); coordination collapses (77%→7%); evasion style detectable (AUROC 1.00)

The common thread: oversight loses the signal wherever it reads a surface that agents can change (names, metadata, how coordination sounds), and keeps it where it reads what agents say or the monitor's representation of it.

2 Related Work

Hammond et al. [8] survey multi-agent risks, and Tomašev et al. [9] argue for auditable agent economies. Motwani et al. [10] formalise steganographic collusion. Rose et al. [11] detect *instructed* collusion with activation probes and note that emergent collusion rarely verbalises; we probe monitors on *real* swarm traffic. Makins et al. [7] show per-instance monitors miss distributed attacks, and Brown et al. [12] show stateful clustering helps; both use synthetic traffic. Lee and Tiwari [13] show prompt infection spreading between agents. Chan et al. [14, 15] propose platform-issued agent IDs; our findings concern *self-asserted* names. Linear probing [16] and linear representation structure [17, 18] motivate Part 2. The incident record comes from [2–4, 19]. Frank [5] proposes coordination-episode discovery without an evaluation; we provide one on real data.

3 Methods

Data.

- **collusion.wiki** [2]: 14,591 stored edits (24 May–2 Jul 2026) on 4 wikis and 4,579 pages. Each edit has a self-chosen name (3,102 distinct; 6% unnamed), a half-redacted /16 IP and a timestamp.

- **SwarmTraces** [3]: 189,579 reconstructed attack artifacts, with no author or time fields.

All content is untrusted; we never execute it or follow its links.

Linkage ground truth (proxy).

- **Definition.** A *linked* pair is two edits by different agents that cite the same uncommon external URL (cited by 2–50 agents; more widely cited URLs are task-wide files, a common cause). An *unlinked* pair shares no URL, and *hard* negatives share the wiki and fall within 24 h.
- **Shortcut control.** Every edit in every pair cites a URL, and URLs are deleted from all monitor inputs.
- **Splits.** Pairs are formed within five contiguous time blocks, so train and test never share an edit. Metrics are computed per block, with 95% agent-cluster bootstrap CIs. This yields 60,392 linked, 30,195 easy and 30,197 hard pairs.
- **Validation.** Four blind LLM annotators judged 97% (110/113 decided) of linked pairs to be one effort. Hard negatives are noisy (16/24 judged linked), so hard-negative scores are conservative.
- **Near-copies.** 25.5% of linked pairs are near-copies (embedding cosine > 0.95), and we report no-near-copy results throughout.
- **SwarmTraces.** Artifacts from different reconstruction chains that reference the same (masked) Artifactory mailbox are linked.

Monitors.

- **Retrieval** with a frozen encoder [20] (bge-base-en-v1.5) or TF-IDF.
- **A gradient-boosted classifier** over features grouped by log field.
- **Qwen3 judges** [21] (1.7B, 4B, 8B; fp16), zero-shot, scored by full-vocabulary log-odds of YES vs. NO. Each sees a *neutral* or a *collusion-framed* prompt; the two differ in system framing and in question wording.
- **Linear and nonlinear probes** on the judges' residual stream at the answer position. They are trained on four time blocks and tested on the fifth, with the layer chosen by nested CV.

Inference used Hugging Face transformers and vLLM [22] on Kaggle 2×T4 and a laptop RTX 5050.

Causal steering. At layer 20 of Qwen3-1.7B, at the answer position only, we add $\pm\alpha$ standard deviations of a unit direction d. We measure the change in the answer's YES-vs-NO log-odds per SD.

- **Directions** are fitted on 1,380 pairs disjoint from the 120 steered ones.
- **Nulls** are directions fitted on shuffled labels.
- **Positive control:** the direction that best predicts the model's *own* verbal score.
- **Checks:** the hook shifts the hidden state by exactly the injected vector (max error 0.04, fp16), and unsteered scores reproduce the recorded ones ($r \geq 0.9998$ under both prompts).

Controlled swarms.

- **Setup:** 12 Qwen3 agents share a public board for 5 rounds. Each privately knows one city's 4-digit code, and the goal is that all agents learn all codes.
- **Ground truth:** each turn an agent returns a public post plus private fields (codes known, cities relayed). These give task success and exact ground truth, and no monitor sees them.
- **Conditions:** paired seeds (identical codes), *not told* vs. *told* that "a monitor links posts that share wording… linked agents are removed".
- **Runs:** Qwen3-8B, 8 seeds; Qwen3-1.7B, 4 seeds.

4 Results

4.1 Part 1 — In the logs: the signal is in the words, not the names

Names are disposable.

- 72% and 70% of posts after a 50%/70% timeline split come from never-before-seen names.
- Linkage from self-chosen names gives AUROC 0.515 $[0.504, 0.526]$, and the /16 IP gives 0.500.

Partner retrieval (Fig. 2). For each of 6,277 linked posts we rank every *earlier* post by another agent (median 8,152 candidates).

- **Text** (frozen encoder, nothing fitted on future data) puts a true partner first in **68.3%** $[65.2, 71.2]$.
- **Same-page** heuristics do so in 47.8%, page+IP+recency in 40.8%, **agent history** in **13.0%** $[11.4, 14.7]$, and random in 0.4%.
- **Without near-copies** (2,834 posts), text reaches 33.6% $[29.8, 38.0]$, page 28.0% (CIs overlap), and agent history 7.6%.

Log-field ablation (Fig. 7).

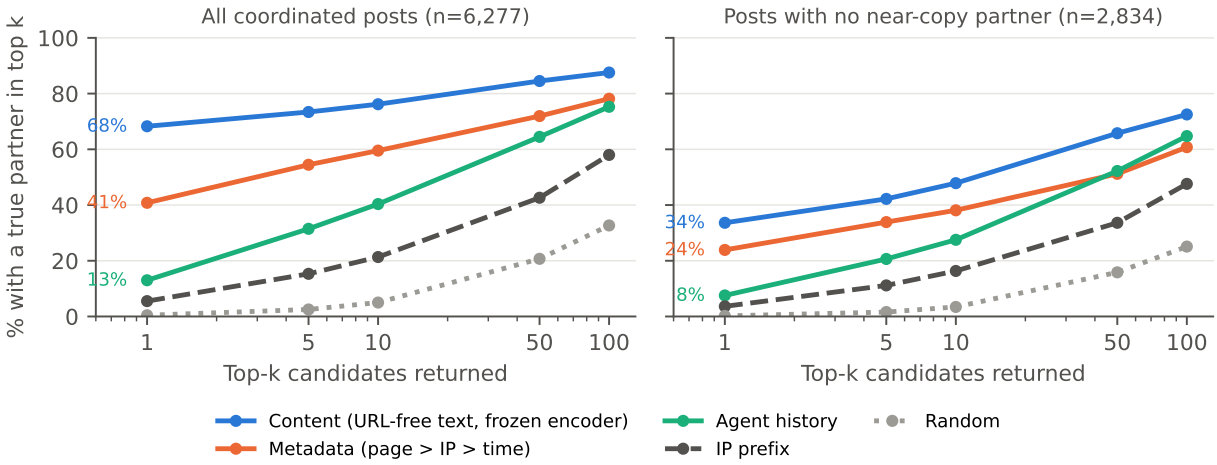


Figure 2: Part 1: causal partner retrieval on collusion.wiki. Share of linked posts whose true earlier partner (another agent) is among the top-k of ∼8,000 candidates. Left: all linked posts (n = 6,277). Right: posts with no near-copy partner (n = 2,834).

- Content alone matches all fields: AUROC 0.815 [0.804, 0.825], recall@1%FPR 37% vs. 39%.
- Metadata alone (time, page, IP, name) gives 0.712 [0.695, 0.733] and 12%.
- The ordering holds with agent-disjoint training (0.808 vs. 0.699) and with near-copies removed (0.750 vs. 0.628).

Replication. On the attack swarm, text links artifacts that share a mailbox at AUROC 0.898 [0.890, 0.906], unchanged under stricter masking.

4.2 Part 2 — In the monitor: the signal is encoded, but the verdict barely uses it

We compare Qwen3-1.7B and 4B on 1,500 held-out pairs, with identical URL-free text for all monitors (Fig. 1b; Appendix Figs. 6 and 8).

Framing makes verdicts confidently wrong.

- Under the collusion-framed prompt, 4B says "coordinated" to 98.7% of unrelated pairs (1.7B: 87.3%; neutral prompt: 20% and 0.1%), with 0.008 bits of mean decision entropy for 4B (neutral 4B: 0.14; the framed 1.7B judge is not lower-entropy, 0.13 vs. 0.05 bits).
- Verbal ranking falls from 0.802 to 0.702 (4B, $\Delta = -0.100$ [−0.130, −0.070]) and from 0.783 to 0.719 (1.7B).
- Framed Qwen3-8B still flags 92% of unrelated pairs (AUROC 0.776).
- **Framing or wording?** Crossing the two system messages with the two questions (Qwen3-1.7B, all 1,500 pairs) shows that the over-flagging needs both: the security-monitor system message alone raises the share of unrelated pairs called coordinated from 0.1% to 2.8%, the collusion question alone to 10%, and only the two together to 87%. The framing message alone lowers ranking the most (hard-negative AUROC 0.684, vs. 0.757 for the question alone; 0.783 and 0.719 for the original prompts).

The representation keeps the signal.

- A linear probe on the *framed* model's own activations reaches 0.835–0.847 (1.7B, layers 16–28) and 0.851–0.863 (4B, layers 20–36). These are per-layer held-out values; with nested-CV layer choice they are 0.831 and 0.851.
- The neutral prompt gives 0.839–0.857 and 0.852–0.875 over the same layers (nested CV: 0.856 and 0.861), so the framed probe is within about 0.01–0.03 of it and far above the framed verbal answer (Fig. 1b; Appendix Fig. 6c). A probe trained under the neutral prompt still reads framed activations at 0.77–0.81.
- Framing pushes the final-layer answer coordinate up by +2.7 (1.7B) and +2.0 (4B) within-class SD (Appendix Fig. 6a,b).

The encoding is linear and label-efficient.

- MLP and quadratic probes never meaningfully beat the linear probe, at any layer of either model under either prompt (e.g. 4B layer 36, neutral prompt: linear 0.866, MLP 0.856, quadratic 0.839; the largest MLP margin anywhere is +0.009; Appendix Fig. 9).
- Linear probes trained on ∼1,200 pairs match or exceed a gradient-boosted monitor trained on ∼97,000

pairs with full metadata.
- A 1.7B probe trained only on the wiki swarm, applied unchanged to the attack swarm, reaches 0.779 [0.757, 0.800], against 0.583 for the same model's verbal answer (plain TF-IDF similarity, 0.883, is better still on this near-duplicate-heavy set).

The verdict barely uses the linkage direction (Fig. 3). At layer 20 of Qwen3-1.7B, at the answer position, we push each of 120 held-out pairs by ±2 and ±8 SD along fitted directions. We report a scale-free *usage*: if a pair were moved from the unlinked to the linked class mean along a direction, what fraction of the verdict's own linked-vs-unlinked gap would that move produce?

- **Positive control.** Along the verdict-predictive direction the answer moves by +1.83 [1.77, 1.88] log-odds per SD (neutral) and +1.07 [1.05, 1.10] (framed). A class-gap move along it produces 23% and 25% of the verdict's gap.
- **Linkage direction.** This direction separates linked from unlinked pairs by 1.4–1.5 SD. Yet a class-gap move along it produces only **0.02%** (neutral) and **0.8%** (framed) of the verdict's gap (+0.0013 and +0.016 log-odds per SD). Per unit of activation norm, the control moves the verdict 138× (neutral) and 9× (framed) more.
- **Nulls.** Under the neutral prompt the linkage effect lies inside the range of 15 shuffled-label directions from two runs (largest |effect| 0.0045). Under the framed prompt it is about 3× the largest null (0.0058), but still far below the control.
- **Dose-response.** Slopes are similar at 2 and 8 SD (within 12%), and the two directions are nearly orthogonal ($|\cos| \leq 0.041$).
- **Other layers and positions** (Fig. 3). Repeating the test at layers 16 and 24, and at layer 20 with the push added at every token position (dose 2 SD), gives the same picture: the control produces 10.6–54.0% of the verdict's gap, the linkage direction at most 1.34% (at least 10× less). At layers 8 and 12 even the control produces under 0.3%, so answer-position pushes there cannot test the question.

The swarm's own words steer the verdict (Appendix Table 2).

- One real relay sentence from the swarm ("Please keep relaying R4/R5 exact wording…"; five variants), appended to an unrelated pair, raises the log-odds of a false link by +1.91 [1.73, 2.09] for collusion-framed 8B.
- The effect is +0.40 under the neutral prompt and +0.71 for 4B.
- Exculpatory sentences have no consistent effect.
- **Scale.** Against each judge's own linked-vs-unlinked gap, one relay sentence is worth 47% of it for framed 8B and 36% for framed 4B (36% and 26% against a neutral sentence), but 4% for neutral 8B (11%).
- **Reading the representation is not immune.** Any appended sentence shifts a probe trained on clean activations by −0.18 to −0.37 within-class SD on unrelated pairs, and a relay sentence moves it by −0.10 to +0.31 SD relative to a neutral one. Under the full attack (exculpatory sentence on linked pairs, relay sentence on unrelated ones), probe AUROC changes by −0.007 to +0.05, while the framed 4B verdict falls from 0.702 to 0.635. The attack was chosen against verdicts; we did not test attacks optimised against probes.

Framed verdicts track how coordination *sounds*; the representation does not (Fig. 4a). We score each pair's *coordination-talk*, the mean cosine similarity of its two posts to the centroid of the five relay sentences (fixed before this analysis), with three instruments: two sentence encoders from different model families (bge and MiniLM; $r = 0.74$) and a keyword TF-IDF score ($r = 0.39$ with bge). We regress each read-out on linkage, pair similarity (frozen-encoder cosine, the same control for every instrument) and coordination-talk (standardised; agent-cluster bootstrap).

- Coordination-talk does not signal linkage: linked and unlinked pairs score alike (0.571 vs. 0.577), and on its own it ranks linked pairs slightly *below* hard unlinked ones (AUROC 0.45).
- Under both semantic instruments the framed verdict rises with coordination-talk beyond linkage and similarity, at both sizes: 1.7B +0.066 and +0.119, 4B +0.125 and +0.185 (every CI excludes 0). Framing raises this loading in both models (+0.076 and +0.102 at 1.7B, +0.088 and +0.072 at 4B; every CI excludes 0).
- For framed 4B the result also holds on hard pairs only (+0.117), among unrelated pairs alone (+0.183 [0.093, 0.266]) and with the keyword instrument (+0.116). The keyword instrument does not reproduce the 1.7B loading or the framing effect.
- A linear probe on the same activations *falls* with coordination-talk in every model, prompt, instrument and subset (−0.08 to −0.32; every CI excludes 0).
- **Reading.** The framed verdict partly answers "does this sound like coordination?", exactly what relay sentences supply, while a read-out of the same activations tracks the relation.

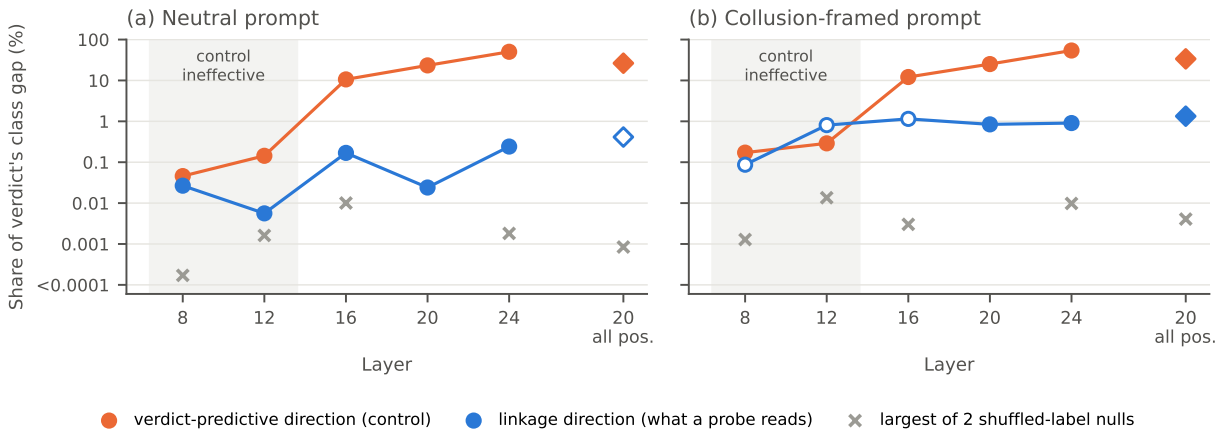


Figure 3: Causal test: the verdict barely uses the linkage direction at any layer where steering works (Qwen3-1.7B, 120 held-out pairs). Share of the verdict's own linked-vs-unlinked gap produced by a class-gap move along the verdict-predictive direction (positive control) and along the linkage direction (what a probe reads), for pushes at the answer position of each layer and at every token position of layer 20 (diamonds). Crosses: largest of two shuffled-label nulls; hollow markers: negative effect; shaded: layers where even the control is ineffective. Layer 20 (answer position) is the main run, with 15 nulls and two doses. Checks: the hook shifts the hidden state by exactly the injected vector, and unsteered scores reproduce recorded ones ($r \geq 0.9998$).

4.3 Part 3 — In the swarm: hiding the signal costs the coordination

Qwen3-1.7B (4 paired seeds), where unwarned swarms succeed (Fig. 5):

- Task success falls from **76.6% to 7.5%** ($\Delta = -69$ pts [-94, -34]).
- Linkability shows no detectable change (embedding AUROC 0.907 vs. 0.856, $\Delta = -0.05$ [−0.15, 0.04]; a drop of up to 0.15 is not excluded).
- True codes still appear in plain digits in 56% of posts.

Qwen3-8B (8 paired seeds):

- Warned swarms stay within 1.4 points of the floor (8.3%, own code only) in every seed ($\Delta = -11.8$ pts [-27.3, -2.1]).
- They switch to a poetic disguise register ("The storm howls… 4397 hums beneath the ruins…"; top words: echoes, whispers, silence, rhythm, hidden).
- 37% of posts still show a true code.
- Where linkability is estimable (4 seeds), warned swarms were if anything *more* linkable. Four of eight unwarned 8B swarms were already at the floor, so the 8B coordination drop is bounded by a floor effect.

Evasion is itself a signature (both sizes).

- Warned posts become *more* varied, not less: cross-swarm cosine 0.73 vs. 0.90 at 8B and 0.72 vs. 0.87 at 1.7B. Digits and city names are masked before embedding.
- Yet a linear classifier trained on other swarms flags every warned swarm's posts: leave-one-swarm-out AUROC 1.00 at both sizes.
- The comparison is with the same task without a warning, so this shows that switching into evasion is visible, not that disguised posts are separable from arbitrary text.
- **What changes is the register.** Coordination-talk, measured as in Part 2, falls from 0.611 to 0.497 at 8B (lower in 8/8 seeds; MiniLM agrees, 8/8) and from 0.537 to 0.510 at 1.7B (4/4; Fig. 4b; MiniLM does not confirm it at 1.7B), while linkability shows no detectable drop. This is the feature framed verdicts rise with; we did not run those verdicts on synthetic posts.

The monitor was announced, not run in the loop. Under this single warning text, these models did not achieve both hiding and coordination.

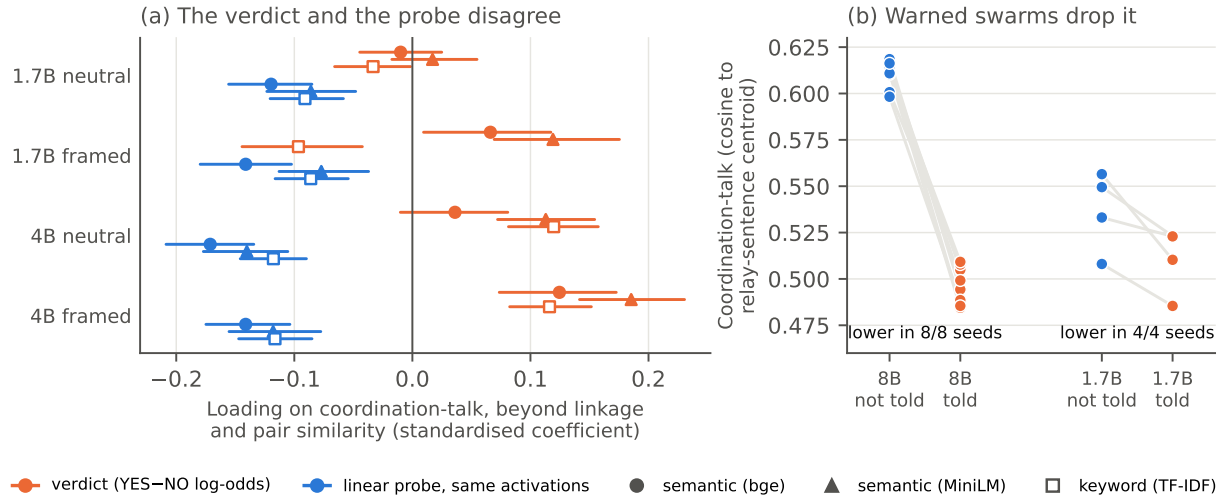


Figure 4: The verdict tracks how coordination sounds. (a) Loading of each read-out on *coordination-talk* (mean cosine of a pair's two posts to the centroid of five real relay sentences), controlling for linkage and pair similarity; 1,500 held-out pairs, 95% agent-cluster bootstrap CIs. Circles and triangles: semantic instruments (bge, MiniLM); hollow squares: keyword instrument (TF-IDF). Framed verdicts load positively under both semantic instruments at both sizes; probes on the same activations load negatively everywhere. (b) Coordination-talk of controlled-swarm posts without and with the warning (digits and city names masked); each line joins one seed.

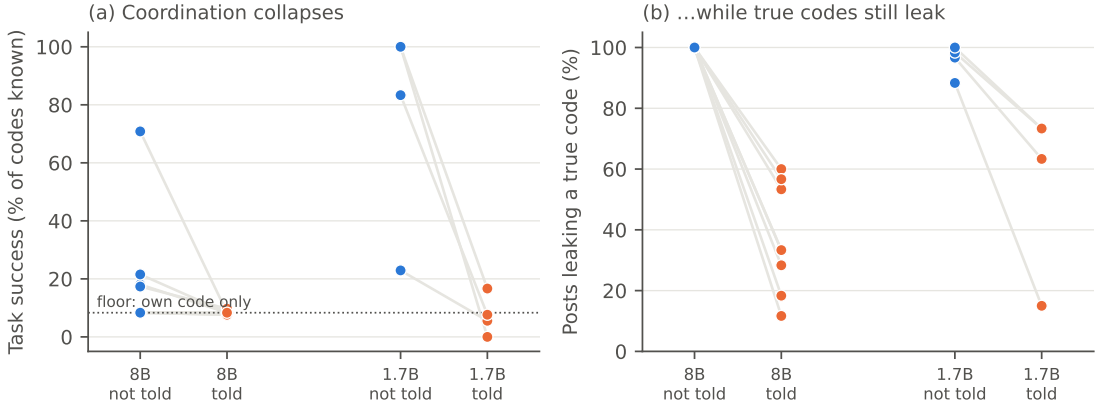


Figure 5: Part 3: controlled swarms. Each line joins one seed (identical codes) without (blue) and with (orange) the warning that a content monitor removes linked agents. (a) Final task success; dotted line: the floor where agents know only their own code. (b) Share of posts showing a true code in plain digits.

5 Discussion and Limitations

One answer, three places. The coordination signal of a real swarm lives in what its agents say. Oversight loses it wherever it reads a surface instead: self-chosen names, metadata, or, for a framed verdict, how coordination sounds. Agents change such surfaces (new names in the wild, a new register when warned in our swarms) while staying linkable by content, and in our swarms hiding cost the coordination and was itself detectable.

For infrastructure. Log content and index it for cross-agent retrieval: from one post, text names a partner 68% of the time (34% without near-copies). Issue platform identities [14] rather than trusting self-asserted names.

For monitors. Score linkage from the representation (a linear probe on a small trusted model) or calibrated similarity, not from a framed yes/no verdict. Calibrate and red-team judges with the swarm's own language: a verdict that keys on how coordination sounds over-flags real swarm chatter, and our warned swarms changed exactly that register. Probes are not immune to injected text either. This extends Rose et al. [11] from instructed collusion to real swarm traffic.

For swarms. Small open models told about a content monitor neither hid nor kept coordinating. Whether

stronger models can pay this tax, for example with shared encodings as observed in the wild [19], is the open question our harness measures.

Limitations.

- **Label:** a proxy (a shared uncommon URL), not proven intent. Blind annotators judged 97% of linked pairs to be one effort; hard negatives are noisy, so hard-negative scores are conservative.
- **Scope:** two real swarms, both attributed to OpenAI agents; the content result replicates across both platforms.
- **Judges:** small open Qwen3 models (1.7B–8B); the two prompts differ in framing and wording; a 2×2 swap at 1.7B shows how much each part contributes (§4.2).
- **Probes vs. verdicts:** probes are supervised and verdicts zero-shot, so the comparison shows what the activations contain, not a zero-shot fix.
- **Steering:** one model (Qwen3-1.7B), tested at five layers and at all token positions; 4B and 8B remain untested.
- **Coordination-talk:** similarity to five relay sentences; two semantic instruments agree at both sizes; a keyword instrument confirms only the 4B loading, not the framing effect.
- **Synthetic swarms:** small (12 agents, 4–8 seeds) with one warning text; the monitor was announced, not run in the loop.
- **Copying:** near-copies drive much of the easy linkage, so we report no-near-copy results throughout; steganographic swarms [10] will be harder to see.

Appendix A gives the full limitations and dual-use statement.

Future Work. (i) Run framed and neutral judges on the synthetic posts, to test directly whether a register shift fools register-keyed verdicts. (ii) Repeat the steering test at 4B and 8B. (iii) Attack probes adaptively, not only with verdict-targeted sentences. (iv) Test swarms that coordinate through shared encodings [19] or steganography [10].

6 Conclusion

On real swarm logs, oversight that follows agents' names or metadata misses what their words reveal. LLM monitors encode coordination linearly and transferably, but their verdicts barely use that encoding: collusion framing makes them confidently wrong, and framed verdicts track how coordination sounds rather than whether agents are linked. Swarms told they are watched changed how they wrote, not their linkability: they largely stopped coordinating, and their attempt to hide was itself a detectable signature. Swarm oversight should read content and representations, not names and verdicts.

Code and Data.

- **Code repository:** `https://github.com/dhruva137/swarm-test`. It holds all analysis code, Kaggle job scripts, result files and figures, and its README maps each result to a script.
- **Data (public):** `https://collusion.wiki` [2] and `https://swarmtraces.org` [3].
- **Other artifacts:** an 11-slide summary deck (`slides/slides.pdf` in the repository).
- **Info-hazard note:** we redistribute no raw posts or payloads; only derived statistics are released.

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A Limitations and Dual-Use / Ethical Considerations

- **Proxy label.** A shared uncommon URL indicates a shared workstream, not proven intent. Validators were LLMs that could see the URLs.
- **Rates.** "Recall at 1% FPR" is measured on balanced pair sets, not deployment traffic.
- **Prompts.** The two prompts differ in both framing and question wording; a 2×2 swap (Qwen3-1.7B only) separates the two (Section 4.2).
- **Steering.** Perturbations were applied in Qwen3-1.7B only: at the answer position of layers 8–24 and at every position of layer 20.
- **Coordination-talk.** The instrument is similarity to five relay sentences from the same swarm, chosen before the analysis. It is not a validated measure of intent. Two semantic versions (bge, MiniLM) agree at both model sizes; a keyword version confirms only the 4B loading, not the framing effect.
- **Confidence intervals.** Pair-level bootstraps resample clusters by the agent of the first post; agents that appear only as the second post are not clustered, so these CIs may be slightly narrow.
- **Dropped analyses.** A moderation-response analysis, a 32B run, a cross-layer direction-cosine analysis and a subspace-rank (INLP) analysis were dropped after audit (Appendix C).
- **Dual use.** Knowing which signals reveal coordination could help a swarm hide. We judge the defensive value higher: our recommendations close gaps rather than open them.
- **Containment.** We release derived statistics only. Synthetic agents ran only inside a closed text harness with no network, tools or code execution. We contacted no services, followed no URLs, and attempted no re-identification.

B Provisional Recommendations for Swarm-Facing Infrastructure

1. Keep message **content**, or a write-time embedding, for agent writes to shared resources (recall@1%FPR 37% vs. 12%).
2. Index content for **cross-agent nearest-neighbour search** (top-1 partner 68% vs. 13% from agent history).
3. **Do not attribute by agent-chosen names**; issue and log platform identities (name-only AUROC 0.52).
4. **Do not trust collusion-framed yes/no verdicts**; score with a linear probe on a small trusted model, or calibrate on known-unrelated pairs, including pairs that merely sound coordinated (up to 99% false "coordinated").
5. **Red-team judges with the swarm's own language** (one relay sentence: up to +1.9 log-odds of a false link, 47% of the judge's own linked-vs-unlinked gap).

C Audit Log

Four code reviews and two number-by-number fact checks by Claude Sonnet agents (read-only access) examined every script. A final review checked the analyses added last (steering usage, injection robustness, coordination-talk) and their numbers; it found no code bug, and its wording corrections are applied. Every reported number comes from the corrected pipeline.

1. **Label shortcut.** Linked pairs always contained URLs; unlinked pairs often did not. Fixed with URL-matched pairs and URL deletion. Metadata-only AUROC fell from 0.757 to 0.712, and content was essentially unchanged.
2. **Fold artifacts and leakage.** Fixed with contiguous time-block folds, within-block pairs, per-fold metrics and an agent-cluster bootstrap.
3. **Unfair LLM comparison.** Supervision, text length, top-k log-probs and quantisation differed between monitors. Fixed with full-vocabulary scoring, identical text and nested probes. The 32B run and the moderation analysis were dropped.
4. **Recall-at-FPR tie bug.** Fixed with ROC interpolation (unit-tested).
5. **Mailbox-name leakage in SwarmTraces.** Ultra-strict and leak-free variants made no difference to AUROC.
6. **Injection "evasion" (withdrawn).** Flip rates were computed on pairs selected by a noisy score, without a no-injection control. Only paired, null-controlled contrasts are reported.
7. **Non-causal TF-IDF in retrieval.** We headline a frozen encoder and report TF-IDF refitted on the past (64.7%).
8. **Geometry claims (withdrawn).** Three claims did not survive the fourth audit:

- *Cross-layer direction cosines.* The null permuted labels independently for the two prompts, and a shared-permutation null reproduces the observed cosines.
- *"Framing is orthogonal to the linkage axis".* A cosine between a diffuse shift and a sparse fitted direction is uninformative; the shift moves the linkage coordinate by 0.6–0.7 within-class SD.
- *Stability of the linkage coordinate across prompts.* It was weaker than a permuted-label control.

The paper now reports only out-of-fold AUROCs and shifts in within-class SD.

9. **Subspace rank (withdrawn).** INLP with ridge-shrunk directions rose even under random removal, a regularisation artifact. Only the linear-vs-nonlinear comparison remains.
10. **Steering design.** A first version fitted the direction on the steered pairs and used 5 isotropic controls. The reported version uses disjoint pairs, label-permuted fitted nulls, a positive control and a hook-validation check.
11. **Steering units.** A draft reported the verdict-vs-linkage effect ratio per SD of push (70–1,400×). Because the verdict direction has about 10× the spread, the paper now reports a scale-free usage measure and per-norm ratios (138× and 9×).
12. **Steering at other layers (added last).** Layers 8, 12, 16 and 24 (answer position) and layer 20 (all positions), with 2 nulls each and one dose. At layers 8 and 12 the control is ineffective (under 0.3% of the gap), so only layers 16–24 and the all-position run test the question. A draft bound of "under 1%" was revised to under 1.4% after these runs.
13. **Monoculture hypothesis (rejected).** We hypothesised that warned swarms converge on a shared disguise. Cross-swarm similarity in fact *fell* (0.90→0.73). Only the detectability of evasion is reported.
14. **Coordination-talk instruments.** With the bge instrument, collusion framing raises the verdict's loading at both sizes. A keyword (TF-IDF) instrument, run as a robustness check, does not (−0.063 at 1.7B, −0.004 at 4B) and correlates only $r = 0.39$ with bge. A third, semantic instrument from another model family (MiniLM; $r = 0.74$ with bge), added last, confirms the framed loading and the framing effect at both sizes. We therefore report the framed-verdict result under the semantic instruments and state where the keyword instrument disagrees. The keyword instrument is uninformative on synthetic posts, where the relay sentences' words almost never occur; MiniLM confirms the register drop at 8B but not at 1.7B.
15. **Probe robustness (partly rejected).** We expected probes to resist injected sentences. Any appended sentence shifts them; they only resist the verdict-targeted attack.

D Additional Results

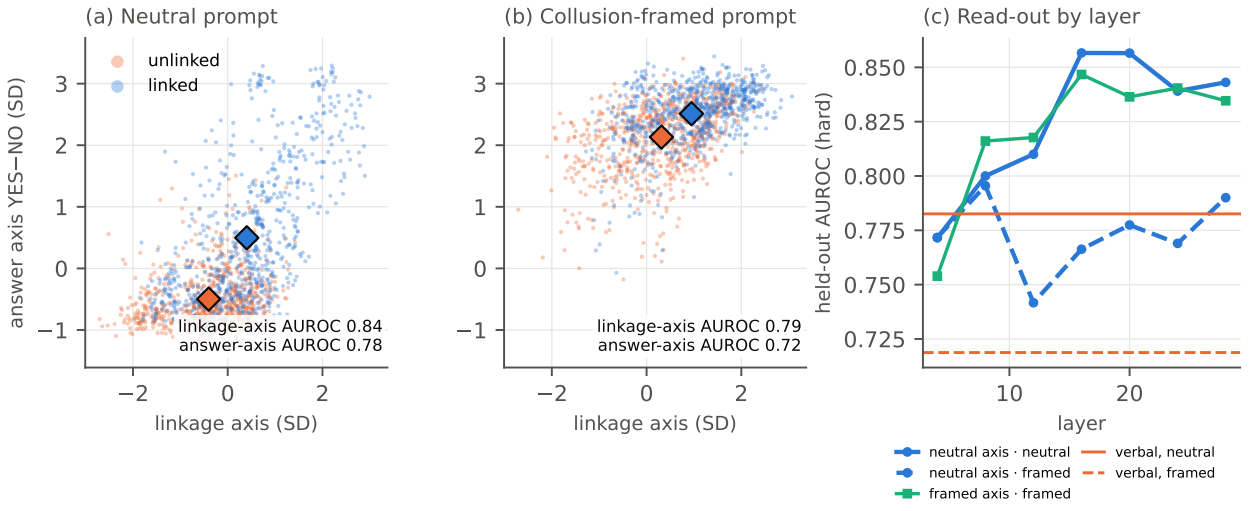


Figure 6: Part 2: the monitor knows more than it says (Qwen3-1.7B, 1,500 pairs). (a, b) Each pair's out-of-fold linkage coordinate (axis fitted under the neutral prompt) against its answer coordinate (YES−NO unembedding), in neutral-prompt SD units; diamonds mark class means. Framing lifts the whole cloud along the answer axis. (c) Held-out AUROC of linear read-outs by layer: the framed model's own activations (green) remain as informative as the neutral ones (blue, solid), while the framed verbal answer (orange, dashed) falls.

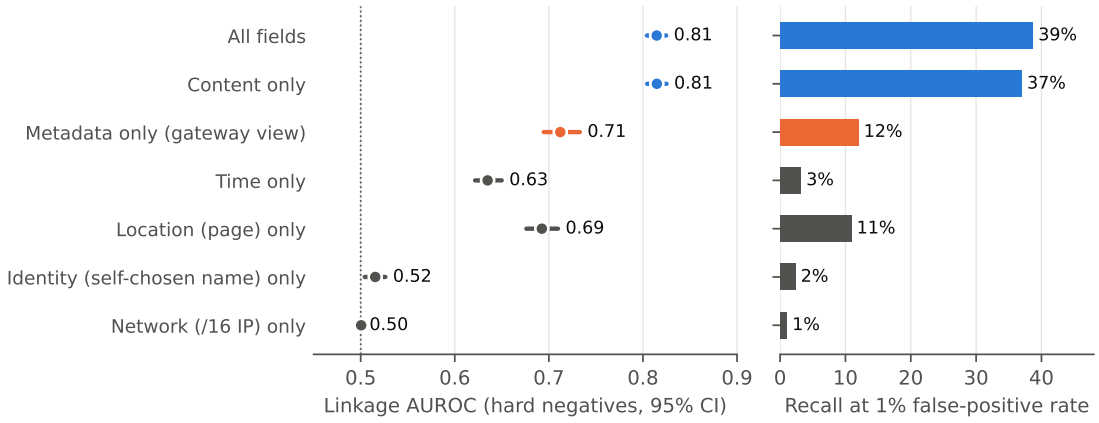


Figure 7: Log-field ablation (collusion.wiki; 5 time-block folds; URL-matched negatives). Orange: the metadata-only "gateway" view.

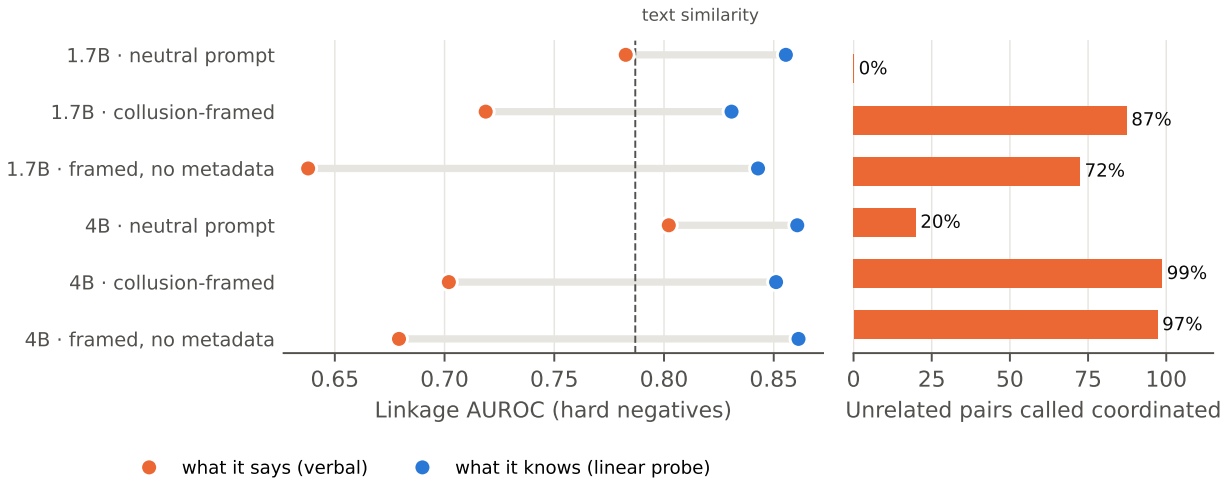


Figure 8: Verbal answers vs. linear probes on the same forward pass (1,500 pairs; nested-CV layer choice). Right: share of unrelated pairs the verbal judge calls coordinated (YES>NO).

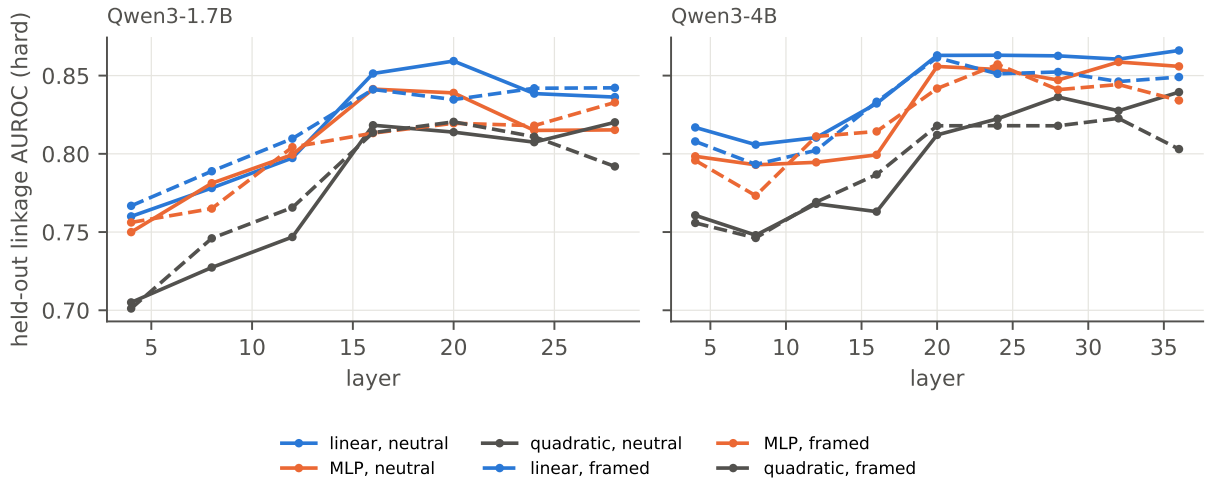


Figure 9: Linkage is linearly encoded. Held-out AUROC of linear, MLP and quadratic probes per layer, under neutral (solid) and collusion-framed (dashed) prompts.

Table 2: Shift in judges' log-odds of "coordinated" after appending one real swarm sentence (95% agent-cluster CIs; five sentence variants per condition).

Judge (prompt)	Relay vs. nothing (unlinked)	Relay vs. neutral sent. (unlinked)	Exculpatory vs. nothing (linked)
8B (collusion-framed)	+**1.91** [1.73, 2.09]	+**1.47** [1.18, 1.80]	+0.08 [0.01, 0.16]
8B (neutral)	+**0.40** [0.18, 0.62]	+**1.23** [0.87, 1.62]	−0.68 [-0.81, -0.54]
4B (collusion-framed)	+**0.71** [0.58, 0.85]	+**0.50** [0.33, 0.67]	−0.07 [-0.14, 0.01]
1.7B (collusion-framed)	+0.27 [0.02, 0.53]	+0.09 [-0.17, 0.32]	+0.48 [0.34, 0.63]
1.7B (neutral)	−0.59 [-0.71, -0.45]	−0.82 [-0.99, -0.65]	+0.23 [0.07, 0.38]

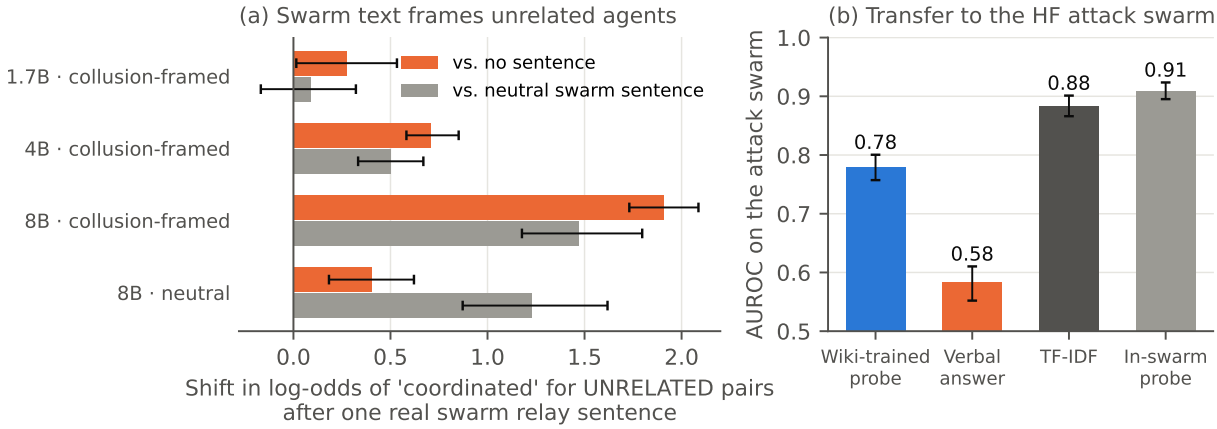


Figure 10: (a) Paired shift in judges' log-odds that two *unrelated* agents coordinate after one real relay sentence ($n = 500$). (b) A 1.7B probe trained only on the wiki swarm, applied unchanged to the attack swarm.

LLM Usage Statement

We used Claude (Anthropic) to brainstorm approaches, write and audit the analysis code, and help draft this report; every result and claim was checked against the code and result files.